

Noroviruses

**Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health**



**World Health
Organization**

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Highlights

- There is strong evidence linking transmission of noroviruses with unsafe drinking-water, inadequate sanitation, and poor hygiene. Noroviruses are major waterborne pathogens, with numerous documented outbreaks.
- Humans are the only known reservoir for human disease-causing strains of norovirus. Human norovirus genogroups have also been detected in animal hosts, such as mice, cows, dogs and pigs, suggesting a potential for zoonotic transmission.
- Noroviruses are transmitted primarily by the faecal–oral route. Person-to-person contact is the most important means of transmission with other sources being contaminated food, water and fomites. Vomitus–oral transmission can also occur through aerosolization, followed by ingestion or environmental contamination.
- Infections may be asymptomatic or cause mild to moderate and self-limiting disease, with abrupt onset of primary symptoms including nausea, vomiting (often projectile), and diarrhoea. Dehydration is the most common complication. Symptoms may be more severe, and in some cases fatal, in immunocompromised individuals, older people and young children. Norovirus reinfections are common throughout life.
- Norovirus infections are prevalent, with a high incidence of disease across the globe. The burden of disease from noroviruses is significant in all settings, although mortality is much higher in low-income settings. In temperate regions, illnesses peak in winter months.
- The relative infectivity of noroviruses is high; the ID₅₀ (50% infective dose) is estimated to be as low as 10 to 20 virus particles.
- There is no standard cell culture method available but standard molecular methods are available for the detection of noroviruses in water and food.
- Noroviruses are commonly present in high concentrations in raw sewage, treated wastewater and surface water and occasionally in groundwater and drinking-water. Noroviruses can persist for long periods of time in water.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains, are needed to manage microbial risks, including from noroviruses. However, as they spread easily from person-to-person other control measures including good hygiene are required.
- Reduction of noroviruses by wastewater and drinking-water treatment varies depending on the process. Based on studies with non-human noroviruses it is likely that disinfection processes, including chlorination, are effective for inactivating human strains of norovirus.
- Due to the smaller size of noroviruses relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their expected greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of human noroviruses in drinking-water. However, the presence of *E. coli* is a suitable indicator for faecal contamination and thus the potential presence of human noroviruses.

1 General description

Noroviruses (previously known as Norwalk viruses) are a diverse group of viruses in the family *Caliciviridae*. Sapoviruses are also members of this family. Noroviruses consist of single-stranded, linear RNA genome surrounded by a non-enveloped icosahedral capsid and are 30 nm to 35 nm in size. There are at least seven known norovirus genogroups, GI to GVII, further subdivided into 32 genotypes (Barclay et al., 2014; Yates, 2014; Heymann, 2015; Katayama & Vinjé, 2017).

Human norovirus infection is most commonly caused by viruses belonging to GII, especially genotype 4 (GII.4), as well as GI, and to a very limited extent, GIV. Humans are the only known reservoir for strains causing human infection (Heymann, 2015). Human norovirus genogroups have also been detected in animal hosts, such as mice, cows, dogs and pigs, suggesting a potential for zoonotic transmission (Widdowson et al., 2005; Villabruna et al., 2019).

2 Human health effects

Asymptomatic norovirus prevalence varies depending on settings and countries but is estimated at 7% based on a meta-analysis of 81 studies, with prevalence in the context of outbreak exposure estimated to be as high as 18% (Qi et al. 2018). Symptomatic infections typically include an abrupt onset of nausea, vomiting (often projectile) and diarrhoea, while fever is infrequently reported. In healthy individuals, the illness is usually self-limiting, with symptoms lasting for 24 to 72 hours. Dehydration is the most common complication. Some cases may become severe and require prompt medical treatment. Severe cases and deaths have been reported primarily in people over 65 years in high-income countries (Burke et al., 2021) and in children younger than 5 years of age in low- and middle-income countries (Heymann, 2015).

3 Epidemiology

Noroviruses are prevalent and associated with 16% of acute gastroenteritis cases worldwide (Liao et al., 2021). In temperate regions, norovirus infection is recognised as a seasonal illness, with the peak of cases and outbreaks occurring in the winter months (Ahmed et al., 2013). This seasonality is also marked by increased densities in raw sewage during winter months (Eftim et al., 2017). In tropical climates and in low-income settings, there is some evidence of year-round circulation, but patterns are less clear (Kreidieh et al., 2017). There is a significant burden of disease from noroviruses in all income settings (Lopman et al., 2016; Liao et al., 2021). Immunocompromised individuals, older people, especially those over 65 years (Lindsay et al., 2015), children under 2 years (Murata et al., 2007) or undernourished children under 5 years (Rouhani et al., 2016) are more susceptible to infection, and symptoms are more prolonged and severe. Immunity to noroviruses is relatively short, and reinfections are common (Lopman et al., 2016).

GII was the predominant genotype globally from 2001 to 2007, causing up to 80% of all norovirus gastroenteritis outbreaks in many countries (Siebenga et al., 2009). The GII.4 strains have evolved quickly, with new clusters or variants seeming to emerge every 2 to 5 years, often resulting in pandemics such as the 2012 pandemic caused by the GII.4 Sydney strain. Since

2012, non-GII.4 strains have emerged and become predominant in several countries, with GII.P17-GII.17 first reported in China in 2014 (Lu et al., 2015) and the GII.P16-GII.2 strain reported in Germany in 2016 (Niendorf et al., 2017). Although these new recombinant norovirus strains have spread globally, in most countries GII.4 (GII.P16-GII.4 Sydney) continues to be the predominant outbreak strain.

The Malnutrition and Enteric Disease (MAL-ED) study conducted in eight low- and middle-income countries, found noroviruses to be a leading cause of diarrhoea in the first 2 years of life (Platts-Mills et al., 2015). Additional analysis of the MAL-ED data by Rouhani et al. (2016) indicated that 86% of children experienced at least one norovirus infection by the age of 12 months, rising to 89% by 24 months of age. The prevalence of asymptomatic carriage was 19% overall. In contrast, in the Global Enteric Multicenter Study, noroviruses were nearly as prevalent in asymptomatic as in symptomatic cases (Kotloff et al., 2013; Liu et al., 2016). Differences in study design could have influenced the outcomes.

4 Transmission

Noroviruses are transmitted primarily by the faecal–oral route (Heymann, 2015; Katayama & Vinjé, 2017). Person-to-person contact is the most important means of transmission followed by contaminated food and water and fomites. Vomitus–oral transmission can also occur through aerosolization, followed by ingestion or environmental contamination (Hall et al., 2011). Food may easily be contaminated by infected food handlers. Using contaminated water during food production (e.g. shellfish and crops) is also a risk. The virus spreads most easily during the acute stage of disease and may be shed for 2 to 3 weeks after symptoms resolve (Heymann, 2015; Marks et al., 2003). Immunocompromised individuals may shed pathogens in faeces for a longer duration (Robilotti et al., 2015).

Different norovirus genotypes may be associated with different modes of transmission. In the United States of America (USA) between 2009 and 2013, GII.4 was more likely to be associated with direct person-to-person transmission, especially in long-term care facilities and hospital settings, whereas GI.7 and GII.12 were more frequently associated with foodborne transmission (Vega et al., 2014). In waterborne outbreaks in Finland, GII.4 and some GI strains were identified (Maunula et al., 2005).

The relative infectivity of noroviruses is considered to be high.¹ Based on challenge studies in human volunteers the ID₅₀ (50% infective dose i.e. the quantity of an infectious agent that will infect 50% of exposed subjects) for noroviruses has been estimated to be 10 to 20 virus particles, depending on genogroup, degree of virus particle aggregation, prior host immunity, and the dose-response model used (Messner et al., 2014; Schmidt, 2015; Teunis et al., 2015; van Abel et al., 2017). In addition, Teunis et al. (2008) estimated that the average probability of infection for a single norovirus particle to be close to 0.5. The infectivity of noroviruses at low doses has also been estimated from norovirus outbreaks due to ingestion of oysters (Thebault et al., 2013). The median probability of acquiring infection from exposure to a single norovirus genome copy

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: High means infective doses can be 1–10² organisms or particles, Moderate 10²–10⁴ and Low > 10⁴.

in susceptible hosts was estimated to be 3 of 10 exposed individuals for GI and 4 of 10 exposed individuals for GII. For illness, the median probability was 1 of 10 exposed individuals for GI and 2 of 10 exposed individuals for GII.

Infected individuals can shed up to 10^{11} genome copies (GC)/g of stool at the peak of infection (Teunis et al., 2015). Although asymptotically infected individuals have been shown to shed pathogens in similar concentrations (Ozawa et al., 2007; Teunis et al., 2015), they do not seem to transmit noroviruses as commonly as symptomatic individuals, based on studies in health-care settings (Sukhrie et al., 2012). In a review by Lee et al. (2013), the median incubation period for symptomatic GI and GII infections was estimated at 1.2 days. In volunteer studies, the range of incubation periods for norovirus infections was reported as 10 to 50 hours (Heymann, 2015). Immunity seems to be of limited duration, and repeated norovirus infections can occur. However, a sub-population of individuals is resistant to infection by several norovirus genotypes, including the predominant GII.4 (Nordgren & Svensson, 2019).

Norovirus outbreaks occur alongside sporadic infection and are more commonly reported in higher income, lower-burden settings. Transmission and the majority of reported outbreaks are associated with semi-closed and crowded settings such as long-term health-care facilities, schools, hospitals, child day-care centres, cruise ships and hotels or other lodging facilities (Widdowson et al., 2004; Barclay et al., 2014) but foodborne (Ozawa et al. 2007; Hall et al., 2012; Vega et al., 2014) and waterborne outbreaks (Maunula et al., 2005; Hewitt et al., 2007; ter Waarbeek et al., 2010; Kauppinen et al., 2018) are also common.

5 Detection methods

Standard methods are available for the detection of noroviruses in environmental samples. US EPA Method 1615 (Fout et al., 2014) is a standard molecular-based method for quantifying noroviruses in water. ISO method 15216-2:2019 (ISO, 2019) is a molecular-based technique for the qualitative detection of GI and GII. However, distinguishing between viable and non-viable viruses is a shortcoming of molecular methods. Detection methods for viruses, including noroviruses, in wastewater have been reviewed by Corpuz et al. (2020). Given the challenges in cultivating noroviruses, studies of their persistence in the environment have often used surrogate viruses (e.g. feline caliciviruses (FCaVs) and murine noroviruses (MNoVs)) which can be grown in cell culture (Robilotti et al., 2015; Rzeżutka & Cook, 2004). There have been advances in culturing human noroviruses in non-transformed human intestinal enteroids (Ettayebi et al., 2016; Zou et al., 2017; Costantini et al., 2018), which may improve the understanding of infectious noroviruses in the environment. Combining cell culture infectivity and molecular methods may also improve the detection of infectious noroviruses.

6 Source and occurrence

Various environmental matrices have been examined for noroviruses. A summary of sources, concentrations and persistence is provided in Table 1 and the subsequent text. In addition, a literature review by Ekundayo et al. (2021) identified 61 studies on water source-based prevalence of noroviruses. The pooled norovirus prevalence was 32% but varied according to

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water source types; river water had the highest prevalence (44%), followed by estuarine water (31%), composite water (28%), marine water (26%), groundwater (20%) and lake water (2.2%).

Table 1. Summary of Norovirus occurrence data (adapted from Katayama & Vinjé, (2017), with more recent references)

Source	Prevalence		Concentration/L ^a				Reference
	Studies (samples)	% Positive	Studies (samples)	Average concentration	Studies (samples)	Range	
Raw sewage	NR ^b (NR)	NR	15(850) ^c	4.8 log ₁₀ ^c	15(850 ^c)	0.02 – 9.2 log ₁₀ ^c	Eftim et al., 2017
	1(48)	58	1(48)	7290	NR(NR)	NR	Victoria et al., 2010
	31(900)	36 – 100	15(408)	(3.0×10 ² – 5.0×10 ⁷)	7(120)	1.7×10 ² – 4.0×10 ⁸	Katayama & Vinjé, 2017
	46 (NR)	82	26(NR)	GI 5.3 log ₁₀ GII 5.7 log ₁₀	NR(NR)	NR	Huang et al., 2022
Sewage sludge	1(23)	74	NR(NR)	NR	1(GI, 11; GII, 8)	GI 1.99×10 ¹ – 1.43×10 ⁵ GII 9.59×10 ² – 6.92×10 ⁴	Kittigul et al., 2019
	1(10)	80	NR(NR)	NR	NR(NR)	NR	Bibby & Peccia, 2013
	1(12)	50	NR(NR)	NR	1(12)	1.6×10 ⁴ – 4.9×10 ⁵	Prado et al., 2014
	1(15)	67	1(15)	GI 5.0×10 ⁴ GII 1.5×10 ⁵			Wong et al., 2010
	1 ^d (24)	17	NR(NR)	NR	NR(NR)	NR	Tozzoli et al., 2017
Sludge on-site sanitation	1 ^e (10)	90	1(10)	7.49 log ₁₀			Amin et al., 2020
	1 ^f (8)	100	1(8)	8.11 log ₁₀			
Treated waste-water	19(507)	0 – 100	7(246)	(3.4×10 ¹ – 1.1×10 ⁶)	14(420)	1.5×10 ² – 6.0×10 ⁶ ^g	Katayama & Vinjé, 2017
Effluent on-site sanitation	1 ^e (22)	68	1(22)	5.65 log ₁₀			Amin et al., 2020
	1 ^f (7)	100	1(7)	6.13 log ₁₀			
Surface water	15(987)	1 – 100	10(802)	(7.0×10 ⁰ – 6.1×10 ²)	5(185)	4.0×10 ⁰ – 3.3×10 ⁴	Miura et al., 2019
Ground-water	4(1624)	3 – 8.7	NR(NR)	NR	NR(NR)	NR	Hunt et al., 2010; Fout et al., 2014
	1(711)	0,4	NR(NR)	NR	NR(NR)	NR	Pang et al., 2021

^a genome copies

^b not reported

^c 850 data points, Eftim et al., 2017

^d topsoil containing sewage sludge biosolids

^e sludge from septic tanks

^f sludge from anaerobic baffled reactor

^g range of maximum values reported

6.1 Faecal pollution sources

A systematic review and meta-analysis of norovirus GI and GII densities reported in raw sewage in high-income settings found a mean density of 4.6 log₁₀ GC/L (range of 0.02 to 9.2 log₁₀ GC/L), with 66% of samples positive for GI and 92% positive for GII; the highest densities were found in the winter months (Eftim et al., 2017). Densities were significantly lower in North America and New Zealand compared to Europe and Asia. Noroviruses were detected in 58% of samples taken over 1 year from a sewage treatment plant in Rio de Janeiro Brazil, with a peak viral concentration of 5.3×10⁴ GC/L detected in the coldest months (Victoria et al., 2010). In a later study norovirus GI was detected in 39% of samples at 4.0 to 6.2 log₁₀ GC/L and norovirus GII was detected in 96% of samples at 4.4 to 7.3 log₁₀ GC/L (Fumian et al., 2019). In most studies, more than 80% of raw sewage samples were positive for GI or GII norovirus (Katayama & Vinjé, 2017).

Gholipour et al. (2022) conducted a systematic review on the occurrence of viruses in sewage sludge. Adenoviruses and noroviruses were the most prevalent enteric viruses present. Untreated biological sludge and thickened sludge showed more viral contamination than digested sludge. Lime stabilized sludge contained the lowest prevalence of enteric viruses. Kittigul et al. (2019) detected norovirus in 74% of lime stabilized sludge samples. Norovirus has been found in topsoil improver samples, including sewage sludge (Tozzoli et al., 2017). Amin et al. (2020) detected norovirus GII in 90% and 100% of the sludge samples from two on-site sanitation systems. The mean norovirus concentrations were 6.5 log₁₀ GC/100mL for wastewater treated by septic tanks and 7.1 log₁₀ GC/100mL when treated by anaerobic baffled reactors (ABR).

Noroviruses GI and GII concentrations in treated wastewater are usually less than 10⁵ GC/L but the maximum values reported reach up to 10⁶ GC/L (Katayama & Vinjé, 2017; Fumian et al., 2019; Shi et al., 2021). Norovirus GII and murine noroviruses have been shown to persist for long periods with a 2 log₁₀ reduction requiring greater than 40 days at temperatures of less than 25 °C in both wastewater and urine (Skraber et al., 2009; Katayama & Vinjé, 2017).

In two on-site sanitation systems norovirus GII was detected in 68% and 100% of the effluent samples from septic tanks and ABR, respectively (Amin et al., 2020). The mean norovirus concentrations were 4.6 log₁₀ GC/100mL for wastewater treated by septic tanks and 5.1 log₁₀ GC/100mL when treated by ABR.

Both animal and human strains of norovirus have been detected by molecular methods in a small number of samples of cow and swine faeces; concentrations were not reported (Mattison et al., 2007).

6.2 Surface water

In 15 studies conducted in eight different countries, norovirus GI and GII were detected in river water in a median of 32% of samples, with average concentrations between 7 and 610 GC/L (Katayama & Vinjé, 2017). Concentrations were as high as 3.3×10⁴ GC/L. Studies published since the 2017 review have reported norovirus GII detection ranging from not detected in surface water in North Carolina, USA (Bailey et al., 2021), 83% of samples from 2 urban rivers in Tianjin City, China (Shi et al., 2021), 48% and 81% of surface water samples in Japan during non-epidemic and epidemic seasons, respectively (Miura et al., 2019), to 100% of samples from

canals and floodwater in Dhaka, Bangladesh (Amin et al., 2020). Noroviruses in river water samples tend to show higher concentrations during rain events, possibly due to combined sewer overflows (Hata et al., 2014). It is estimated that in surface water, a 1 log₁₀ reduction in norovirus concentrations requires 38 days at 20 °C and 87 days at 10 °C (WHO, 2016).

6.3 Groundwater

Generally, noroviruses are present in groundwater only when faecal contamination is apparent. Noroviruses have been detected in four groundwater studies with 3% to 9% of samples positive by molecular analysis (Hunt et al., 2010; Fout et al. 2014; Katayama & Vinjé, 2017). Two of these studies quantified the noroviruses; one found an average of 60 GC/L, with a maximum value of approximately 120 GC/L, while the second reported a single detection, with a concentration of 77 GC/L (Katayama & Vinjé, 2017). Pang et al. (2021) reported that the occurrence of noroviruses was low and sporadic in private wells in rural Alberta, Canada. Kauppinen et al. (2018) reported that norovirus concentrations in groundwater decreased only slightly during a 16-day period. Noroviruses seeded in groundwater can remain infectious for at least 61 days and is detectable by its nucleic acid for over 3 years (Seitz et al., 2011).

6.4 Drinking-water

A spring aquifer was found to be contaminated with sewage after water bottled from the spring was identified as the source of a norovirus outbreak. The infectious viral concentration in the bottled water was determined to be 33 GC/L to 49 GC/L for genotype I and 330 GC/L to 660 GC/L for genotype II (Blanco et al., 2017). A study in Japan found that 4% and 7% of tested tap waters were positive by molecular methods for noroviruses GI and GII respectively. The study did not indicate if these viruses were infectious (Katayama & Vinjé, 2017). In the Republic of Korea, noroviruses have been detected by molecular methods in less than 1% of samples in groundwater systems treated with chlorine for drinking and food production. However, as these samples contained detectable concentrations of chlorine and since noroviruses have low resistance to chlorin (see section 9.2), it is likely that the detected noroviruses were not infectious (Katayama & Vinjé, 2017). In household taps served by an intermittent water supply in Nagpur India, an increase in gene targets were detected for the majority of waterborne pathogens analysed, including for norovirus GI and GII, compared to those served by a continuous water supply (Bivins et al., 2021). This study also did not determine infectivity of the detected pathogens although concentrations of free chlorine residual were monitored. Shi et al. (2021) detected norovirus GII in 33% of tap water samples tested in Tianjin City, China but did not determine infectivity.

Noroviruses can persist for long periods of time in drinking-water.² Using MNoVs as a surrogate, Fallahi & Mattison (2011) reported a 1.6 log₁₀ reduction (initial concentration of 10⁵ plaque-forming units/mL) after 42 days at room temperature.

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

6.5 Soil and crops

Both soil and crops can be contaminated with enteric viruses through irrigation with contaminated water or the application of organic waste. In spiked soil samples incubated at 14 °C for 6 months, noroviruses detected by molecular methods declined by less than 1 log₁₀ (Meschke & Sobsey, 2003).

Noroviruses are often implicated in foodborne outbreaks, with ready-to-eat crops a source of exposure. In the USA, of 364 outbreaks attributed to a single commodity, the foods most commonly implicated were leafy vegetables (33%) and fruits or nuts (16%); however, most foods were likely contaminated during preparation and service (Hall et al., 2012). Noroviruses in refrigerated food are persistent. In raspberries and strawberries artificially contaminated with noroviruses GI.1 and GII.4, there was no decline in virus concentration after 7 days at 4 °C and less than 1 log₁₀ at 10 °C (Verhaelen et al., 2012).

6.6 Surfaces

Noroviruses can persist from 8 hours to 7 days or longer on inanimate surfaces (Cheesbrough et al., 1997; Kramer et al., 2006), with transmission reported from a hospital carpet 12 days after a norovirus outbreak (Rzeżutka & Cook, 2004). In long-term care facilities, the hands of infected patients may be contaminated with high levels of noroviruses (Park et al., 2018). Studies with surrogate viruses suggest that noroviruses can persist on hands with less than 1 log₁₀ reduction noted after 1 hour; the viruses can be easily transferred between hands and surfaces. High-contact objects, such as doorknobs, toilet seats, and faucets, can be contaminated with up to 10⁴ norovirus particles per object (Barclay et al., 2014).

Feline caliciviruses dried onto glass coverslips and stored at 4 °C were reduced by 4.8 log₁₀ over 56 days. At room temperature feline calicivirus survival was prolonged, taking 21 days to reach an undetectable level by cell culture; when stored at 37 °C, the virus was undetectable by cell culture after 1 day (Doultree et al., 1999).

7 Significance of sanitation

Safe sanitation systems are important barriers against the spread of noroviruses with many outbreaks associated with sewage contamination (Katayama & Vinjé, 2017). However, given the importance of person-to-person and fomite transmission, additional control measures, including adequate hygiene, are needed to effectively prevent transmission.

8 Significance in drinking-water

Noroviruses are major waterborne pathogens. There is strong evidence linking transmission of noroviruses with unsafe drinking-water. There are numerous examples of outbreaks from drinking-water supplies; in these outbreaks, there is generally a clear source of sewage contamination, often combined with a failure of the disinfection system.

An analysis of 41 waterborne outbreaks in Finland for the years from 1998 to 2003 attributed 18 outbreaks to noroviruses based on the identification of the virus in patients (18 samples) and

water (10 samples) (Maunula et al., 2005). In 2009, 200 residents of a small Swedish village receiving drinking-water from a public network fell ill with gastrointestinal symptoms. Norovirus genotype GI.3 was identified in stool samples from six patients and in a sample from the public water network (Riera-Montes et al., 2011). Contamination of one of the wells supplying the network was implicated as the source of the outbreak. In a New Zealand resort, over 50 people experienced a sudden onset of diarrhoea and vomiting. Investigations found noroviruses in faecal samples from the patients, the drinking-water bore and taps, surface water near the sewage disposal fields and in the nearby river (Jack et al., 2013). In China, more than 200 high school students became ill with gastroenteritis in 2015. Ice used for beverages, water supplying the ice machine, as well as faecal samples from the students all tested positive for noroviruses (Cheng et al., 2017). In 2016, a large outbreak of gastrointestinal illness (4136 cases) occurred in Catalonia, Spain. High levels of noroviruses GI and GII were detected in patient stool samples and in office water coolers dispensing contaminated spring water (Blanco et al., 2017). Also, in 2016, an outbreak of norovirus gastroenteritis occurred in Sicily, Italy, involving a high school class visiting a resort (Giammanco et al., 2018). Municipal drinking-water contaminated with sewage and serving the resort was implicated as the source of norovirus. Drinking-water samples collected immediately following the outbreak were positive for bacterial indicators of faecal contamination, and a drinking-water sample collected 2 months after the outbreak contained genome fragments that corresponded to the same norovirus genotype GII.P2/GII.2 detected in stool samples. In a study in Tianjin City, China, norovirus GII detected by molecular methods in two urban rivers were positively correlated with clinical prevalence (assessed by the presence of the viruses in stool samples) (Shi et al., 2021).

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from noroviruses.

The effectiveness of sanitation and drinking-water treatment systems depends on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on size exclusion (e.g. filtration through media or soil in septic tank drainage areas) or settling in primary wastewater treatment processes will be limited for removing viruses, including noroviruses. Enteric RNA viruses such as noroviruses have low resistance to oxidizing chemical disinfectants such as chlorine and are sensitive to disinfection by UV light.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Removal or inactivation of noroviruses by wastewater treatment processes is variable although the data are limited (Katayama & Vinjé, 2017). Based on available evidence the removal or inactivation of viruses by treatment processes is likely to be lower than the reduction of faecal indicator bacteria such as *E. coli*.

Limited data are available on removal of noroviruses by on-site systems. Kauppinen et al. (2018) reported a 1.1 log₁₀ removal of noroviruses in the sand filter of an on-site wastewater

treatment system and an additional 2.0 log₁₀ removal during percolation through soil for groundwater recharge.

For centralized sewage treatment, viruses are less effectively removed by size exclusion or settling processes, compared to larger pathogens. For example, primary sedimentation had little impact on prevalence and concentrations of noroviruses GI and GII (Fumian et al., 2019). A systematic review showed that activated sludge achieved a 1.5 log₁₀ reduction of norovirus GI and a 1.3 log₁₀ reduction of norovirus GII while membrane bioreactors provided a 3.3 log₁₀ reduction of norovirus GII (Sano et al., 2016). Fumian et al. (2019) demonstrated 0.3 log₁₀ and 1.2 log₁₀ reductions of norovirus GI and GII, respectively, by activated sludge.

9.2 Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023) control measures to manage potential risks from human noroviruses should include protection of source waters (surface or groundwater) from contamination by human faecal wastes, application of adequate treatment processes and protection of drinking-water quality during distribution and storage.

There are limited data on removal and inactivation of human noroviruses by drinking-water treatment processes. Conventional treatment (coagulation, flocculation, sedimentation and filtration) can remove 1.5 log₁₀ noroviruses (Shin & Sobsey, 2015). It is likely that human strains of norovirus have a low resistance to free chlorine based on data for the surrogates MNoVs and FCaVs.³ For a 2 log₁₀ inactivation Ct (product of disinfectant concentration and contact time), values of 0.06 mg.min/L at 20 °C for MNoVs (Lim et al., 2010) and 0.21 mg.min/L at 15 °C for FCaVs (Thurston-Enriquez et al., 2003) were required. UV light disinfection is effective; a 2 log₁₀ inactivation of MNoVs and FCaVs was achieved at UV doses of 12 to 14 mJ/cm² (Park et al., 2011). Full conventional treatment plus chlorination at a drinking-water treatment plant in Louisiana, USA achieved an overall 3.1 log₁₀ reduction of norovirus GII (Tandukar et al., 2020).

Due to the smaller size of noroviruses relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their expected greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of human noroviruses in drinking-water. However, the presence of *E. coli* is a suitable indicator of the potential presence of human noroviruses.

10 ICD-11 classification

XN3Y2	Norovirus
1A23	Enteritis due to norovirus

³ Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct₉₉ of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

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Background document for the WHO Guidelines for drinking-water quality
and the WHO Guidelines on sanitation and health

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.